

# Anne-Marie George

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I work on preference learning, computational social choice, and fairness in AI — developing methods for eliciting, aggregating, and reasoning about preferences, with an eye on how these choices shape fairness in collective decision-making. My work is primarily theoretical, drawing on combinatorial optimisation, graph theory, logic, and learning theory.

I am a Principal Investigator at Integreat, Norway's Centre for Trustworthy AI, on the Fairness & Explainability theme, and at TRUST — Centre for Trustworthy AI, in the research area *AI for Flourishing*.

## Research

### Preferences & collective decision-making

Methods for learning, eliciting, and aggregating preferences, with applications spanning recommender systems and language model alignment.

### Fairness in AI

How fairness should be conceptualised in preference-based AI systems, and what that means for collective decision-making in practice.

### Methods

Primarily theoretical work drawing on combinatorial optimisation, graph theory, logic, learning theory, and online learning — as well as computational social choice, approval voting, liquid democracy, stable matching, and preference modelling more broadly.

## Teaching & Service

### Teaching

**Lecturer — Logiske metoder**
IN1150, University of Oslo, Spring 2025

**Lecturer — Reinforcement Learning and Decision Making Under Uncertainty**
IN-STK5100 / IN-STK9100, University of Oslo, Spring 2022 and 2023

**Lecturer — Adaptive Methods for Data-Based Decisions**
IN-STK5000 / IN-STK9000, University of Oslo, Autumn 2022

**Lecturer — Mallows Model Tutorial**
Preference Statistics Summer School, July 2025

**Lecturer — Reinforcement Learning Tutorial**
23rd Geilo Winter School on Computational Statistics, January 2023

### PhD Co-Supervision

Lorenzo Zuccato (since 2024)

Meirav Segal (PhD completed 2025)

Thomas Kleine Buening (PhD completed 2024)

### Master's Supervision

“Actionable Recourse Over Time”, 2025/26

“Preference Models for Decision Making: The Moral Machine Case”, 2023/24

“College Admissions with Quotas and Bonus Points”, 2022/23

### Talks & Outreach

“Fairness by Design — Considering Fairness in Algorithmic Decision Making”, Data Science Day, dScience Centre, University of Oslo, 2024

### Service

Local organising committee, International Workshop on Statistical Modelling (IWSM), Oslo 2026

Reviewer / PC member: AAAI, AISTATS, IJCAI, SCALA

Member of the Green Council, Department of Informatics, University of Oslo (since 2023)

### Awards

Outstanding Paper Award, ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO) 2022

Best Paper Award, Cooperative AI Workshop @ NeurIPS 2021

Best Poster Award, COST IC1205 Summer School on Computational Social Choice, 2016

## Publications

**2026**

**Prognostic biomarker discovery in pancreatic cancer through hybrid ensemble feature selection and multi-omics data** JOURNAL

John Zobolas, **Anne-Marie George**, Alberto López, Sebastian Fischer 0004, Marc Becker, Tero Aittokallio
*BioData Min.*

**Cycles in Liquid Democracy: A Game-Theoretic Justification** CONFERENCE

Markus Brill, Rachael Colley, **Anne-Marie George**, Grzegorz Lisowski, Georgios Papasotiropoulos, Ulrike Schmidt-Kraepelin
*IJCAI/ECAI*

**Probably Correct Optimal Stable Matching under Two-Sided Uncertainty** PREPRINT

Andreas Athanasopoulos, **Anne-Marie George**, Christos Dimitrakakis
*CoRR*

**2025**

**Probably Correct Optimal Stable Matching for Two-Sided Market Under Uncertainty** CONFERENCE

Andreas Athanasopoulos, **Anne-Marie George**, Christos Dimitrakakis
*AAMAS*

**On Middle Grounds for Preference Statements** CONFERENCE

**Anne-Marie George**, Ana Ozaki
*IJCAI*

**Probably Correct Optimal Stable Matching for Two-Sided Markets Under Uncertainty** PREPRINT

Andreas Nikolaos Athanasopoulos, **Anne-Marie George**, Christos Dimitrakakis
*CoRR*

**Optimizing Prognostic Biomarker Discovery in Pancreatic Cancer Through Hybrid Ensemble Feature Selection and Multi-Omics Data** PREPRINT
John Zobolas, **Anne-Marie George**, Alberto López, Sebastian Fischer 0004, Marc Becker, Tero Aittokallio
*CoRR*

**2024**

**Minimal Macro-Based Rewritings of Formal Languages: Theory and Applications in Ontology Engineering (and Beyond)** CONFERENCE
Christian Kindermann, **Anne-Marie George**, Bijan Parsia, Uli Sattler
*AAAI*

**Eliciting Kemeny Rankings** CONFERENCE
**Anne-Marie George**, Christos Dimitrakakis
*AAAI*

**Threshold Recourse for Dynamic Allocations** CONFERENCE
Meirav Segal, **Anne-Marie George**, Christos Dimitrakakis
*EWAF*

**Better Luck Next Time: About Robust Recourse in Binary Allocation Problems** CONFERENCE
Meirav Segal, **Anne-Marie George**, Ingrid Chieh Yu, Christos Dimitrakakis
*xAI (3)*

**2023**

**Policy Fairness and Unknown Bias Dynamics in Sequential Allocations** CONFERENCE
Meirav Segal, **Anne-Marie George**, Christos Dimitrakakis
*EAAMO*

**2022**

**The excess method: a multiwinner approval voting procedure to allocate wasted votes** JOURNAL
Steven J. Brams, Markus Brill, **Anne-Marie George**
*Soc. Choice Welf.*

**Liquid Democracy with Ranked Delegations** CONFERENCE
Markus Brill, Théo Delemazure, **Anne-Marie George**, Martin Lackner, Ulrike Schmidt-Kraepelin
*AAAI*

**On Meritocracy in Optimal Set Selection** CONFERENCE
Thomas Kleine Buening, Meirav Segal, Debabrota Basu, **Anne-Marie George**, Christos Dimitrakakis
*EAAMO*

**Interactive Inverse Reinforcement Learning for Cooperative Games** CONFERENCE
Thomas Kleine Büning, **Anne-Marie George**, Christos Dimitrakakis
*ICML*

**Single-Peaked Opinion Updates** CONFERENCE
Robert Bredereck, **Anne-Marie George**, Jonas Israel, Leon Kellerhals
*IJCAI*

**Simulating University Application Data for Fair Matchings** CONFERENCE
Meirav Segal, **Anne-Marie George**, Christos Dimitrakakis
*NAIS*

**2020**

**Assigning and Scheduling Service Visits in a Mixed Urban/Rural Setting** JOURNAL
Mark Antunes, Vincent Armant, Kenneth N. Brown, Daniel A. Desmond, Guillaume Escamocher, **Anne-Marie George**, Diarmuid Grimes, Mike O'Keeffe, Yiqing Lin, Barry O'Sullivan, Cemalettin Ozturk, Luis Quesada 0001, Mohamed Siala 0002, Helmut Simonis, Nic Wilson
*Int. J. Artif. Intell. Tools*

**2019**

**Preference inference based on lexicographic and Pareto models** THESIS
**Anne-Marie George**
*University College Cork, Republic of Ireland*

**2017**

**Efficient Inference and Computation of Optimal Alternatives for Preference Languages Based On Lexicographic Models** CONFERENCE
Nic Wilson, **Anne-Marie George**
*IJCAI*

**Preference Inference Based on Hierarchical and Simple Lexiographic Models** JOURNAL
Nic Wilson, **Anne-Marie George**, Barry O'Sullivan
*FLAP*

**2016**

**Towards Fast Algorithms for the Preference Consistency Problem Based on Hierarchical Models** CONFERENCE
**Anne-Marie George**, Nic Wilson, Barry O'Sullivan
*IJCAI*

**Preference Inference Based on Pareto Models** CONFERENCE
**Anne-Marie George**, Nic Wilson
*SUM*

**2015**

**Computation and Complexity of Preference Inference Based on Hierarchical Models** CONFERENCE
Nic Wilson, **Anne-Marie George**, Barry O'Sullivan
*IJCAI*

**The Comparison of Multi-objective Preference Inference Based on Lexicographic and Weighted Average Models** CONFERENCE
**Anne-Marie George**, Abdul Razak, Nic Wilson
*ICTAI*